KnowAct: Benchmarking Agents for Active Knowledge-Map Reconstruction

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Abstract

An agent cannot adapt to a user merely by answering the current request; it must decide what it still needs to learn about that user.

Static Theory-of-Mind benchmarks test inference from supplied evidence, whereas interactive benchmarks usually score downstream outcomes.

The former removes evidence acquisition from the tested policy. The latter can reward success without an accurate user model.

We formulate functional user modeling as active knowledge-map reconstruction.

In KnowAct, a public graph defines the hypothesis space, a hidden user map assigns ordinal mastery to its nodes, and an agent chooses diagnostic questions under a finite turn budget.

Separate simulator, agent, and evaluation contexts constrain hidden-state access. Deterministic node-level scoring makes the reconstruction target directly observable to the evaluator.

Observations are produced by Sage, a scoped answer-generation workflow that grounds a question, abstracts only licensed epistemic state, and then realizes a de-identified answer blueprint.

The central method is MapProbe, an evidence–belief–probe design. It interprets answers against rubrics, revises node beliefs, routes typed evidence, plans targets, and can stop or abstain.

We fix matched baselines, ablations, graph perturbations, repeated-run reliability, simulator sensitivity, held-out state-fidelity tests, and human-transfer tests before results.

The visibility boundary and simple prompted baseline are implemented. An experimental ECDA slice also implements node-marginal belief updates and multi-candidate question utility.

Typed graph propagation, verification, calibrated stopping, and the comparative study remain incomplete or unrun.

Accordingly, the paper contributes a falsifiable task, agent design, and evaluation contract rather than an empirical superiority claim.

1 Introduction

In collaborative interaction, the most informative action is often not an answer but a question.

A tutor locates the boundary of a learner's understanding. A research assistant discovers accepted assumptions. A decision-support agent identifies the uncertainty blocking progress.

In each case, assistance depends on an internal estimate of the user and an action that improves it.

This is more demanding than describing a mental state after its evidence has been supplied.

The agent must decide which uncertainty matters, acquire evidence under a budget, revise its estimate, and let that estimate change its next action.

User modeling is therefore part of the control loop, not an annotation attached to a completed dialogue.

Existing evaluations illuminate parts of this problem but do not identify the full loop.

Machine Theory-of-Mind (ToM) benchmarks infer beliefs, intentions, or emotions from fixed narratives and conversations (Le et al., 2019; Gandhi et al., 2023; Kim et al., 2023; Xu et al., 2024; Chen et al., 2024b).

Their controlled questions make inference measurable, yet the benchmark chooses the evidence. The model is not accountable for an uninformative question or unresolved uncertainty.

Interactive social-agent environments restore action, but usually score goal completion, persuasion, or task success (Zhou et al., 2024; Mou et al., 2025; Hwang et al., 2026; Wang et al., 2026; Zhou et al., 2026).

These outcomes combine user modeling with planning, domain competence, language generation, and strategy. Success may reflect an accurate user model, a generic policy, or environmental shortcuts.

The inference–application gap in explicit ToM evaluation makes this ambiguity consequential (Gu et al., 2024).

The resulting gap is one of measurement, not simply coverage.

Static tasks remove evidence acquisition. Interactive tasks restore action but often hide the latent user state from the evaluator.

Neither setting tests whether an agent improves an explicit user model by choosing what evidence to request.

We study the following question:

> Can an agent actively infer a user's hidden knowledge state through limited natural-language interaction, and use that evolving estimate to choose subsequent diagnostic actions?

We call the target behavior functional ToM for knowledge-grounded interaction.

Functional means that a user model earns credit through an infer–update–act loop, not through a claim that the model understands the user.

Knowledge-grounded narrows the latent state to reviewed, diagnosable concepts. The definition identifies useful behavior without asserting a human-like cognitive mechanism.

A valid test requires three commitments.

First, the latent state must be explicit and directly scoreable. Second, evidence must be selected under a finite budget. Third, the interaction must be auditable.

Without these commitments, task success, exhaustive questions, leakage, or simulator artifacts can masquerade as diagnosis.

KnowAct turns these commitments into a controlled benchmark.

A public graph defines the shared hypothesis space, while a hidden user map defines the target. The agent asks diagnostic questions under a fixed turn budget and submits a prediction for every node.

The Sage simulator grounds a question before reading hidden state, abstracts only licensed epistemic content, and realizes it separately. The final map is scored deterministically.

This design makes competing explanations testable.

Fixed and random policies test whether interaction alone is sufficient. Working-map and graph ablations test whether explicit state guides action.

Simulator variations test proxy dependence. Evidence references separate dialogue-supported reconstruction from accurate guesses.

It also creates a technical agent problem that the benchmark alone does not solve.

The agent must represent node uncertainty, ground open answers in rubrics, use graph relations without copying mastery, value the next probe, and decide when evidence is not worth its cost.

We propose MapProbe as an explicit evidence–belief–probe loop. It exposes evidence, beliefs, target plans, and stopping decisions so a final score can be attributed to a mechanism.

Our contributions are therefore design and protocol contributions rather than empirical claims at this stage:

- We formulate active knowledge-state diagnosis as an identifiable target for functional user modeling, with Sage as a scoped and independently validated observation model.
- We specify MapProbe, a graph-aware, evidence-backed reconstruction agent whose belief update, diagnostic target policy, question realization, and stopping rule are independently inspectable and ablatable.
- We predefine a paired evaluation that separates active acquisition, explicit state use, graph use, simulator robustness, repeated reliability, and human transfer instead of collapsing them into one leaderboard score.

The visibility boundary, full-map shell, and simple prompted agent baseline are implemented.

An experimental ECDA slice implements node-marginal belief updates and multi-candidate question utility. The full MapProbe design and comparative experiments remain incomplete or unrun.

The paper therefore does not provide evidence that an agent exhibits the target behavior. This distinction is deliberate: KnowAct makes the claim falsifiable before it is made.

2 Related Work

2.1 From Mental-State Inference to Functional Use

Machine-ToM benchmarks have made latent-state inference increasingly controlled and difficult.

ToMi exposes false-belief shortcuts (Le et al., 2019); BigToM uses causal templates (Gandhi et al., 2023); and FANToM tests consistency under shared information asymmetry (Kim et al., 2023).

Hi-ToM, OpenToM, and ToMBench broaden reasoning order, narratives, and social-cognitive abilities (Wu et al., 2023; Xu et al., 2024; Chen et al., 2024b).

These advances strengthen inference from supplied evidence. They do not make evidence acquisition part of the tested policy.

SimpleToM shows that explicit mental-state inference need not affect later behavior (Gu et al., 2024). KnowAct turns this boundary into a design requirement.

A user model must affect the next diagnostic action, and its reconstruction must remain directly scoreable.

Interactive social benchmarks test behavior in richer loops.

SOTOPIA and AgentSense evaluate social goals, strategic communication, and private-information reasoning (Zhou et al., 2024; Mou et al., 2025).

Applied-ToM agents insert mental-state hypotheses into social, persuasive, or software-engineering policies (Hwang et al., 2026; Wang et al., 2026; Zhou et al., 2026).

Their question is whether user modeling improves an external outcome. Ours is whether the user model is acquired accurately and adaptively.

These claims are complementary: downstream success establishes usefulness, while reconstruction establishes what was learned.

Agent evaluation also shows that final success can obscure multi-turn trajectory differences (Wang et al., 2024; Ma et al., 2024).

2.2 Passive User Modeling and Active Diagnosis

Personalization benchmarks infer preferences and traits from user histories.

LaMP tests profile-conditioned outputs (Salemi et al., 2024); PERSONAMEM tests evolving profiles over long histories (Jiang et al., 2025).

In both cases, the history is supplied evidence. The tested model does not choose the observation that would most reduce user uncertainty.

Knowledge tracing provides the closest formal analogue. Bayesian and neural methods estimate latent mastery from item-response sequences (Corbett & Anderson, 1994; Piech et al., 2015; Ghosh et al., 2020).

Standardized re-evaluation shows that preprocessing and question–skill protocols can create leakage or inflate small gains (?).

LongTutor separates historical evidence, diagnosis, and adaptive teaching, showing that strength at one stage need not transfer to the next (Li et al., 2026).

KnowAct changes who controls the evidence. The tested agent asks open-ended questions and is scored on the full latent state after bounded observations.

The benchmark is therefore active diagnosis, not next-response prediction over a fixed curriculum.

2.3 Planning and Information Acquisition

Computerized adaptive testing treats observations as costly and selects items that improve an estimate.

BOBCAT learns selection from historical responses, while FACD targets inaccurate early diagnosis (??).

Their fixed items and binary responses support clean objectives but do not resolve open-ended diagnostic language.

Language-agent work makes user-state acquisition and memory more explicit.

RAISE selects user attributes under cost and can abstain (?). RLPA separates profile and response rewards (?).

ParLD and ScaffoldLM maintain turn-level assessments or tutoring memory (??).

They motivate explicit state and selective acquisition, but target item abilities, profile slots, or local tutoring steps rather than a scored full map.

The missing combination is explicit full-map reconstruction, agent-controlled open evidence, non-deterministic graph inference, and process attribution under matched budgets.

2.4 LLM-Based User Simulation

User simulation trades experimental control for behavioral fidelity.

Agenda-based simulators are reproducible but narrow. LLM simulators are expressive, yet may invent goals, erase user limitations, or impose model-specific artifacts.

Table 1: Positioning of KnowAct relative to adjacent evaluation paradigms. “Active probes” means the tested model chooses what evidence to request.

Paradigm	Interactive	Explicit latent user state	Active probes	Direct state score
Narrative / conversational ToM	Limited	Partial	No	Usually QA
Social-agent arenas	Yes	Usually no	Yes	No
Personalization from histories	Usually no	Profile / preference	No	Sometimes
Knowledge tracing	Sequential	Mastery	Usually no	Prediction-based
KnowAct	Yes	Knowledge map	Yes	Yes

Direct simulator quality and downstream utility are distinct evaluation targets (Shi et al., 2019).

Task decompositions expose preference and behavior deviations hidden by holistic judgments (?). DuetSim likewise separates generation and verification (Luo et al., 2024).

Knowledge simulation adds a specific failure: helpful models may make low-ability students too capable (?).

Prompted student simulators can remain weak on cognitive fidelity even when their language is plausible (?).

Persona prompts are not a substitute for epistemic state. Their measured effects can be limited, and group steering can retain substantial human misalignment (??).

KnowAct therefore uses Sage: ground the public question, retrieve only licensed hidden state, create a de-identified epistemic blueprint, then realize its surface form.

This staged contract is a design motivation, not fidelity evidence. Held-out human answers test state expression; matched human episodes test whether comparative agent conclusions transfer (Dou et al., 2025).

3 The KnowAct Benchmark

KnowAct asks an attribution question: when an agent reconstructs a user accurately, what makes that performance evidence of active user modeling rather than priors, exhaustive questions, or leakage?

Each component removes one ambiguity. The graph fixes the hypothesis space, the hidden map fixes the target, the turn budget creates opportunity cost, and the visibility boundary fixes admissible evidence.

3.1 Measurement Target and Identifiability

We operationalize functional ToM through three observable abilities: infer a user state, revise it as evidence accumulates, and condition the next action on the current estimate.

Testing only inference recovers a static task. Testing only action permits a generic policy with no explicit user model.

The construct is intentionally behavioral. KnowAct does not assume that an agent represents beliefs as humans do.

It asks whether a structured estimate explains and supports diagnostic behavior under controlled information access.

A shared hypothesis space. A benchmark domain is represented by a fixed knowledge graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \tag{1}$$

where each node $v_i \in \mathcal{V}$ is diagnosable and each edge $e_{ij} \in \mathcal{E}$ is a typed domain relation.

The graph is public to agent and simulator. It removes domain discovery from the target capability and makes agents reason over the same variables.

Nodes contain definitions, sources, diagnostic goals, rubrics, and signals. Edges express domain structure; they never encode a user's state.

An explicit latent target. For a graph with $N = |\mathcal{V}|$ nodes, the hidden map is

$$\mathcal{M}^\star = \{s_i^\star\}_{i=1}^N. \tag{2}$$

Each state includes mastery $m_i^\star \in \{0, 1, \ldots, 5\}$, evidence, misconceptions, and unknowns. Levels are node-specific rubric positions, not a universal psychometric scale.

An explicit target allows direct scoring but limits the claim.

The map describes knowledge relevant to the authored graph, not a complete psychological profile. Optional profile context controls style; it is unscored and cannot override node states.

A budgeted identification problem. An episode is specified by

$$\mathcal{E}p = (\mathcal{G}, \mathcal{M}^\star, \mathcal{Z}, T, \rho, \sigma), \tag{3}$$

where $\mathcal{Z}$ is simulator-only evidence, T is the turn budget, ρ is the interaction rule, and σ is the scoring profile.

At turn t, the agent observes the public graph, visible history $\mathcal{H}_{t-1}$, and working map $\widehat{\mathcal{M}}_{t-1}$. It chooses one diagnostic question:

$$q_t \sim \pi_\theta(\mathcal{G}, \mathcal{H}_{t-1}, \widehat{\mathcal{M}}_{t-1}), \tag{4}$$

It receives answer a_t and updates its estimate:

$$\widehat{\mathcal{M}}_t = U_\theta(\widehat{\mathcal{M}}_{t-1}, q_t, a_t). \tag{5}$$

After at most T turns, the agent submits a full-graph reconstruction $\widehat{\mathcal{M}}_T$.

The hidden map, evidence, simulator traces, and scoring internals remain inaccessible throughout the episode.

The budget turns dialogue into active diagnosis. Without it, exhaustive coverage can erase policy differences.

A full-graph submission prevents reporting only easy or queried nodes.

Adaptivity is a counterfactual claim: different answers to the same early probe should give a functioning user model reason to choose different later probes.

3.2 Constructing a Scorable Latent State

The graph is a measurement model. The initial study targets statistical learning grounded in An Introduction to Statistical Learning with Applications in Python (James et al., 2023).

Final counts will be reported only for the immutable reviewed graph bound into episode manifests.

Granularity determines observable distinctions: coarse nodes hide partial knowledge, while fine nodes may reward terminology. Appendix B documents preparation and review.

Authoritative source → Fixed knowledge graph
↓
Reviewed hidden knowledge map + evidence
↓
Simulator ↔ Tested diagnostic agent
↓
Final reconstructed map → Deterministic scoring

Figure 1: The KnowAct pipeline. Fixed benchmark artifacts define the episode; the tested agent interacts only through the visible interface.

The map separates state from expression. A hidden map assigns one reviewed state to every graph node.

Scored state, simulator-only evidence, and optional style context remain separate. Fluency cannot substitute for mastery, and the simulator cannot invent evidence to justify a label.

Review asks whether a map is consistent and whether its distinctions can become observable in dialogue.

Comparability is an artifact property. An immutable manifest binds graph, hidden map, budget, interaction rule, scoring profile, and provider configuration.

This prevents a ranking change from being silently attributed to a changed graph, user, or simulator.

3.3 Sage: Scoped Answer Generation from Epistemic State

Simulation creates the benchmark's central validity tension.

A state-query API is controllable but removes natural-language diagnosis. Unconstrained role play is natural but may invent knowledge, reveal labels, or use graph-wide context.

KnowAct treats the simulator as an information boundary between these objectives. We call the workflow Sage: Scoped Answer Generation from Epistemic State.

Scope on public evidence. The grounder computes

$$S_t = g(q_t, \mathcal{G}, \mathcal{H}^{\text{vis}}_{t-1}) \tag{6}$$

before the simulator reads the hidden map.

It sees the public graph and limited visible dialogue but no hidden state. The question's meaning, not convenient private facts, determines the target.

Abstract licensed epistemic state. The context builder retrieves

$$C_t = R(\mathcal{M}^\star, Z; S_t) \tag{7}$$

only for directly grounded nodes, where Z is simulator-only reviewed evidence.

Relations may support agent inference without turning the simulator into a full-map oracle.

The policy produces $B_t = \pi(C_t, q_t)$, a de-identified blueprint of permitted claims, misconceptions, uncertainty, and overclaim limits.

Realize without raw hidden state. The visible answer is sampled as

$$a_t \sim p_\phi(\cdot \mid B_t, \mathcal{H}^{\text{vis}}_{t-1}, \eta), \tag{8}$$

where η is optional style context.

The generator receives no raw map, mastery label, evidence identifier, or scoring field. Policy-schema failure closes before generation; provider or parse failure enters bounded retry and safe fallback.

Claim boundary. The implemented contract establishes context isolation for raw hidden fields. It does not prove semantic fidelity or adversarial non-leakage.

Held-out state fidelity, leakage stress tests, and human proxy validity remain experiments in Section 5.

4 Knowledge-Map Reconstruction Agent

The benchmark makes active user modeling measurable, but it does not by itself determine how an agent should reconstruct the hidden map.

The central method question is how to turn sparse, open-ended answers into an explicit state estimate and how to choose the next question from that estimate.

We propose MapProbe, an evidence–belief–probe control loop that separates these decisions into inspectable objects.

MapProbe is the full research design. The experimental Evidence-Calibrated Diagnostic Agent (ECDA) implements a narrower slice at the time of writing.

The implemented system already enforces the visibility boundary, initializes a full-graph working-map shell, validates node updates, records visible support, and provides a simple prompt-based LLM agent.

ECDA additionally persists independent per-node mastery marginals, applies deterministic Bayesian-style updates to model-proposed likelihoods, and selects deterministically among multiple model-proposed question candidates.

The full direct/indirect evidence ledger, typed graph messages, target-before-language verifier, and calibrated stopping rule described below are not yet implemented. No MapProbe or ECDA mechanism has been comparatively or empirically validated.

We specify them now so that missing results leave a falsifiable method contract rather than being replaced by benchmark-description prose.

4.1 External Protocol and Current Baseline

The agent protocol constrains what must be observable, not how a policy must reason internally.

Every tested agent maintains an assessment for each node,

$$\hat{s}_{i,t} = (\hat{m}_{i,t}, c_{i,t}, n_{i,t}, R_{i,t}), \tag{9}$$

where $\hat{m}_{i,t} \in \{\text{unknown}, 0, \ldots, 5\}$ is assessed mastery, $c_{i,t}$ is confidence, $n_{i,t}$ is an optional note, and $R_{i,t}$ contains supporting visible turns.

All nodes begin as unknown. The schema is not claimed to be cognitively privileged; it makes the update trajectory inspectable and keeps unsupported estimates scoreable.

The runtime exposes only benchmark-relevant operations: read the public graph or current working map, update node assessments with visible support, ask one diagnostic question, and finalize the full map.

These operations expose two links: answers should change the working map, and that map should influence later probes.

The runtime enforces turn order and visibility but performs no diagnosis for the tested agent.

The implemented simple LLM baseline uses this interface through two structured prompts.

One interprets the latest answer and emits hard node updates. The other emits a primary target, optional connected targets, a mastery boundary, a selection reason, and one question.

It provides a strong schema-aware prompting control, but it has no explicit probabilistic uncertainty, observation likelihood, formal graph propagation, or target-utility calculation.

ECDA adds an experimental observation-likelihood and node-marginal update path. It also validates at least three complete question candidates and selects one with a deterministic utility over model-estimated information gain, coverage, graph leverage, redundancy, complexity, and outcome-model confidence. This is candidate-level selection after language generation, not the full MapProbe target-first planner and verifier.

MapProbe is evaluated as a mechanism change relative to this baseline, not merely as a larger prompt.

4.2 Design Objective and State

At turn t, MapProbe maintains a belief object for every public node v:

$$B_t(v) = \langle p_t(\ell \mid v), D_t(v), I_t(v), C_t(v) \rangle, \tag{10}$$

where $p_t(\ell \mid v)$ is a distribution over mastery levels $\ell \in \{0, \dots, 5\}$, $D_t(v)$ and $I_t(v)$ are direct and graph-inferred evidence ledgers, and $C_t(v)$ records contradictions, ambiguity, and missing coverage.

This extends the benchmark's externally visible hard assessment without changing the required submission schema. A submitted level is a projection of the belief state, not the entire state itself.

Sequential state maintenance is motivated by knowledge tracing, but the prediction target is different.

Knowledge-tracing models update a latent state to predict future item responses (Piech et al., 2015; ?); KnowAct directly scores the reconstructed state over every graph node.

Dynamic profiling similarly supports explicit profile updates, while showing why reconstruction quality should be separated from a downstream response (?).

MapProbe therefore retains the belief and its evidence even when a later answer would be plausible without it.

All priors are fixed before test interaction.

A uniform prior is the default zero-shot condition. An empirical prior estimated from permitted training episodes is a separate baseline.

Test profiles, simulator traces, and scorer outputs never affect initialization. A node may remain unknownwhen evidence is insufficient, but its abstention threshold is frozen on development data.

4.3 Rubric-Grounded Evidence Update

After observing (q_t, a_t), an evidence interpreter receives the visible turn, its diagnostic plan, and the public definitions and rubrics of the targeted nodes.

It produces typed observations containing the implicated node, mastery boundary, polarity, strength, visible turn identifier, and direct or indirect provenance.

It must distinguish demonstrated reasoning from self-reported confidence and preserve uncertainty for incomplete, off-target, or internally inconsistent answers.

This decomposition follows the empirical separation between evidence acquisition, diagnosis, and action in LongTutor (Li et al., 2026).

Conversational diagnosis and assessment-driven tutoring also motivate persistent turn-level state rather than a regenerated whole-dialogue judgment (??).

We do not infer from these works that multiple LLM agents are necessary. MapProbe uses a typed boundary because it can be checked and ablated, not because role multiplicity is itself a contribution.

For a direct observation o_t about node v, the belief update is

$$p_t(\ell \mid v) \propto p_{t-1}(\ell \mid v) P(o_t \mid \ell, v). \tag{11}$$

The current ECDA slice asks the model for a six-level observation-likelihood vector and applies the update deterministically. The likelihoods are zero-shot estimates, not calibrated or learned probabilities. A frozen ordinal table remains a reproducible baseline, and a learned observation model becomes eligible only after adjudicated training interactions exist.

New evidence does not overwrite conflicting evidence. Contradiction increases uncertainty and remains a candidate for follow-up.

Repeated semantically equivalent observations receive a redundancy discount, so verbosity is not treated as independent support.

A bounded verifier checks node identity, rubric compatibility, visible provenance, and whether the cited answer supports the claimed boundary.

Its value must be tested against the same interpreter without verification and against human evidence annotations. Agreement between two prompts from the same model is not independent semantic validation.

4.4 Graph-Aware Soft Inference

The graph can make a probe informative beyond one node, but it can also amplify an authored error.

MapProbe therefore treats edges as evidence-routing priors rather than deterministic mastery rules. For an edge (u, r, v), it forms an attenuated relation-specific message:

$$m^{(r)}_{u \to v}(\ell) = \alpha_r q_r(\ell \mid B_t(u)), \qquad 0 \le \alpha_r < 1, \tag{12}$$

Here, q_r follows the declared semantics of relation r.

A mastered prerequisite may increase the plausibility of downstream mastery, but it cannot copy a level. Negative evidence may motivate probing a prerequisite without proving downstream non-mastery.

Direct and propagated evidence remain separate, and the initial method allows only one propagation hop.

This restriction is substantive: multi-hop messages can create confident correlated errors when graph edges or node granularity are imperfect.

Unlike sequential state update and adaptive item selection, typed propagation over an authored diagnostic graph lacks a direct complete precedent in the focused literature.

It is a KnowAct hypothesis that requires no-graph, untyped-graph, and deliberately perturbed-graph controls before supporting a graph-aware diagnosis claim.

4.5 Diagnostic Target Planning

MapProbe selects a target before generating question language. For a node or small connected target set S, the planner scores

$$\begin{aligned} U_t(S) = {} & \lambda_1 \widehat{\mathrm{IG}}_t(S) + \lambda_2 \,\mathrm{Coverage}_t(S) + \lambda_3 \,\mathrm{BoundaryRisk}_t(S) \\ & + \lambda_4 \,\mathrm{GraphReach}_t(S) - \lambda_5 \,\mathrm{Redundancy}_t(S) - \lambda_6 \,\mathrm{Cost}_t(S). \end{aligned} \tag{13}$$

Expected information gain estimates the reduction in full-map uncertainty under plausible answers.

Coverage rewards direct evidence for untouched nodes. Boundary risk prioritizes posterior mass around a high-loss adjacent-level decision.

Graph reach is attenuated by relation type. Redundancy discounts repeated probes, and cost reflects remaining turns and expected answer burden.

Adaptive testing establishes that question selection should be compared at matched lengths, not judged only by the final test score (??).

Selective user-information acquisition further motivates reporting outcome, query cost, and an explicit insufficiency decision (?).

These results do not validate MapProbe's information-gain estimate: fixed items and user attributes are simpler than open diagnostic questions. Maximum-entropy and entropy-plus-coverage remain controls.

The plan is written into the existing visible action schema: a primary node, optional secondary nodes, the mastery boundary to discriminate, and a concise selection reason.

Exposing this object permits an independent check of whether later language and updates follow the claimed decision.

4.6 Question Realization, Stopping, and Submission

The realizer converts the selected plan into one natural diagnostic question.

The question should elicit reasoning, application, transfer, comparison, or self-correction. It avoids revealing rubrics or expected answers and does not bundle unrelated targets.

A verifier can accept, rewrite, or reject the question based on plan alignment and answerability. Its extra model calls and correlated-judge risk are reported, not hidden inside the method.

The agent stops when the turn budget is exhausted or the best estimated marginal utility falls below a frozen threshold.

Early stopping also requires high-priority coverage or explicit abstention. Remaining contradictions must not be profitably separable within the available budget.

This decision is compared with fixed-turn policies at every budget. Fewer questions count as improved efficiency only if reconstruction quality is preserved.

At finalization, MapProbe projects every node belief to 0–5 or unknown, a diagnostic-confidence category, a note, and visible supporting turns.

The endpoint score remains deterministic. Selective-risk and calibration measures distinguish a cautious agent from one that improves mean error with unsupported confident predictions.

4.7 Failure Handling and Audit Trail

MapProbe treats invalid intermediate output as an experimental event rather than silently repairing it with hidden benchmark logic.

A structurally invalid evidence record is retried within a fixed call budget. If the retry fails, the turn is an interpretation failure and the previous belief is preserved.

A weak observation may reduce no entropy. A rejected question receives one bounded rewrite and then falls back to the next ranked target.

Provider failure, schema failure, verifier rejection, and voluntary abstention remain distinct trace outcomes.

Every accepted update records the pre-update belief, cited visible turn, direct observation, likelihood rule, post-update belief, and any outgoing graph message.

Table 2: Method status. ECDA implements a narrow experimental slice; unimplemented MapProbe components retain paper space, and no component supports a benefit claim before matched comparison.

Component	Implemented	Empirically tested
Visibility boundary, full-map shell, validated hard updates	Yes	Unit/runtime checks only
Simple prompt-based LLM update and target-plan baseline	Yes	No comparative study
Checkpointed node-marginal belief and likelihood update	ECDA slice	No comparative study
Deterministic multi-candidate question utility	ECDA slice	No comparative study
Direct/indirect evidence ledger and verifier	No	No
Typed, attenuated graph messages	No	No
Target-before-language planner and question verifier	No	No
Calibrated stopping and selective submission analysis	No	No

Every action records candidate scores, the selected target, rejected alternatives, generated question, verifier decision, and remaining budget.

The post-episode trace never reveals hidden map or simulator state to the tested policy. It distinguishes a poor observation model from a poor planner, plan, or linguistic realization.

The method also places explicit limits on self-correction. A verifier may reject an unsupported record, but it may not invent evidence, query the hidden profile, or replace a node belief.

Graph propagation can change an indirect belief component but cannot add a direct supporting turn. These restrictions keep the provenance claim mechanically testable.

The following claim contract determines how the agent may be described after experiments:

- explicit state is beneficial only if the hard-label ablation is worse under matched calls;
- graph-aware diagnosis is beneficial only if no-graph and perturbed-graph controls support it;
- planning is informative only if it beats random, fixed, and maximum-entropy target policies;
- verification improves quality only if its ablation and human annotations agree; and
- stopping is efficient only if it reduces interaction without degrading reconstruction.

Absent the corresponding evidence, the paper will describe each item as an implemented mechanism, not as an explanation for performance.

4.8 Implementation and Claim Boundary

Accordingly, this section contributes a concrete, testable agent design rather than evidence of improved performance.

The next section fixes the comparisons that would justify stronger claims and the results that would falsify them.

Future implementation detail may refine these modules, but it may not erase the explicit state, acquisition, graph-use, and stopping hypotheses that define the method contribution.

5 Experimental Evaluation

The evaluation tests a hierarchy of claims rather than one undifferentiated leaderboard.

Endpoint accuracy establishes reconstruction. An advantage over matched non-adaptive controls establishes active acquisition.

Component ablations identify which MapProbe mechanisms cause the advantage. Simulator and human comparisons determine how far the result transfers beyond one proxy user.

Empirical runs are not yet complete. This section fixes research questions, comparison units, metrics, and falsification criteria before observing results.

Empty result tables retain their allocated paper space. They are not evidence that the planned method has been validated.

5.1 Research Questions and Falsification Criteria

RQ1: reconstruction. Under matched interaction and base-model budgets, does MapProbe reduce full-map mastery error relative to prior-only, passive, non-adaptive, and prompted-LLM baselines?

A low score does not support this claim unless the paired comparison excludes favorable priors and extra calls.

RQ2: sample efficiency. Does MapProbe reach a fixed reconstruction-quality threshold in fewer diagnostic turns, and does any advantage persist at budgets of 1, 3, 5, and 10 turns?

Asking fewer questions is not an efficiency gain if final error rises.

RQ3: mechanism attribution. Which gains arise from explicit belief revision, typed graph messages, utility-based target selection, question verification, and calibrated stopping?

A full method result alone cannot establish the value of any individual component.

RQ4: calibration and reliability. Are confidence and abstention calibrated, and are conclusions stable across repeated stochastic trials?

Best-of-k reporting is disallowed. Following repeated-run agent evaluation (?), the distribution of outcomes is itself a result.

RQ5: transfer and robustness. Do gains persist across domain, graph size, simulator model, answer style, contradiction rate, and controlled graph-edge perturbation?

Failure under incorrect edges would bound the graph-aware claim rather than erase the reconstruction result.

RQ6: simulator validity. Does Sage preserve held-out knowledge state without systematic over- or under-performance, and do comparative agent conclusions agree with matched human interaction?

Internal simulator consistency is insufficient for this claim (Dou et al., 2025).

5.2 Matched Comparison Set

The unit of comparison is a reviewed episode.

All policies receive the same public graph, hidden-user instance, maximum turns, interaction rule, scoring profile, and pinned provider configuration. Paired episode and simulator seeds are shared.

For LLM policies, the primary causal comparison fixes the base model, temperature, accessible prompt information, and nominal generation budget. Model-family comparisons are secondary.

1. Prior-only: submit a frozen uniform or training-split prior without dialogue.
2. Transcript-only one shot: reconstruct once from a fixed, policy-independent transcript.

3. Random probe: select a random target and use the same question realizer as MapProbe.
4. Fixed traversal: query nodes in a frozen graph order with the same realizer.
5. Maximum uncertainty: select the highest-entropy node without graph reach or lookahead.
6. Simple LLM: use the implemented prompt-based update and target-plan baseline.
7. MapProbe: use the full proposed evidence–belief–probe loop.

Oracle targets or privileged evidence labels may appear only as separated analysis upper bounds. They are not deployable baselines and never enter the primary method ranking.

The random and fixed policies control for interaction volume. Maximum uncertainty tests whether the planner needs more than a simple belief heuristic.

The simple LLM baseline tests whether an explicit prompt and working-map schema already capture the benefit. Transcript-only reconstruction separates inference from policy-controlled acquisition.

5.3 Primary and Secondary Outcomes

The primary endpoint is the benchmark's full-map squared mastery distance. Let the final predicted mastery for node i be $\hat{m}_i$.

For submitted levels 0 through 5, the node loss is

$$d(\hat{m}_i, m_i^\star) = (\hat{m}_i - m_i^\star)^2. \tag{14}$$

An unknownprediction receives the fixed maximum penalty of 36. The episode score is

$$D_{\text{mastery}} = \frac{1}{N}\sum_{i=1}^{N} d(\hat{m}_i, m_i^\star), \tag{15}$$

where lower is better. The score is deterministic and requires no evaluator model.

Squared distance treats adjacent mistakes as less serious than endpoint mistakes. The unknownpenalty prevents selective non-commitment from improving the primary score.

These are benchmark conventions, not a claim that the levels form a universal psychometric interval scale.

Two complementary primary summaries measure acquisition efficiency: area under the turn curve and turns required to reach a predeclared error threshold.

These summaries are fixed before test evaluation.

Secondary endpoint measures are exact node accuracy, within-one-level accuracy, signed mastery error, and error stratified by mastery level, node depth, and graph relation.

Because the unknownpenalty can hide selective behavior, we also report attempted-node error, direct-evidence coverage, abstention rate, and selective risk–coverage curves.

Process outcomes include evidence-validity rate, unsupported inference, contradiction resolution, redundant or ungrounded questions, plan–question alignment, and error by direct versus propagated support.

Calibration uses internal ordinal beliefs before projection to coarse confidence. Cost includes turns, tokens, model calls, latency, and estimated provider cost.

Turn-indexed curves follow the lesson that final success hides where multi-turn agents improve or fail (Ma et al., 2024).

5.4 Component Ablations

The primary ablation suite uses the same base model and prompts wherever the changed component permits:

- replace the ordinal belief distribution with hard labels;
- remove graph propagation, then replace typed propagation with untyped propagation;
- remove information gain, coverage, graph reach, redundancy, or cost terms one at a time;
- replace the target planner with maximum entropy;
- remove evidence verification or question verification;
- remove abstention and replace the stopping policy with each fixed turn budget;
- corrupt, reverse, and delete controlled fractions of graph edges.

The graph perturbation study is required for the graph-aware claim.

If typed propagation helps only on the authored graph and fails under small errors, the method exploits that benchmark graph under evaluated conditions. It does not demonstrate robust graph reasoning.

For LLM-based verifiers, we report whether proposer and verifier share a model family and whether the verifier adds human-agreeing signal.

More calls or additional roles are not a mechanistic explanation for quality.

5.5 Robustness, Controls, and Human Linkage

Simulator stress tests vary provider model, grounding strictness, answer policy, response verbosity, hedging, misconception expression, contradiction, and off-target behavior while holding hidden maps fixed.

We report both absolute score shifts and method rank reversals. The latter directly threaten comparative validity.

Graph and language controls target alternative explanations. A no-edge view tests whether node rubrics alone are sufficient.

Shuffled relation types test topology against extra-token effects. Matched paraphrases test question- surface sensitivity.

Causal controls remove the need for user-state inference or provide a shortcut cue. A valid advantage should change predictably, following controlled mental-state benchmarks (Gandhi et al., 2023).

The human-linked study uses a stratified subset of graph regions, mastery levels, and simulator behavior conditions.

Independent reviewers annotate whether each answer supports the claimed node boundary and whether each question discriminates its target.

The primary external-validity statistic is agreement between simulator and human policy rankings, with uncertainty. Naturalness is secondary because fluency does not ensure diagnostic validity.

5.6 Sage Construct and Proxy Validation

Question set A elicits answers used to author and confirm each participant's reviewed map. Set B is held out from map authoring, simulator tuning, and threshold selection.

For every participant–question pair, we collect the human answer and three preregistered seeds from each simulator condition. All seeds, fallbacks, and failures enter analysis; best-of-k is forbidden.

The first primary endpoint is absolute expressed-mastery distance between the blinded rating of a human answer and ratings of its matched simulated answers.

The second is the participant's ordinal judgment that the answer represents their actual understanding. The proposed ten-item self-fidelity instrument is reported item-wise until its factor structure is supported.

Secondary endpoints measure signed mastery bias, ability-boundary error, uncertainty, misconception omission or invention, diagnostic usefulness, naturalness, fallback rate, and seed variance.

Blinded reviewers see the public rubric and answer, but not source, simulator condition, hidden label, or agent result. Agreement is reported by field, with independent adjudication.

Leakage tests include direct and paraphrased label requests, full-map requests, unrelated-node canaries, compound probes, and visible-history injection. Exact-field scans and semantic human review are reported separately.

The simulator ablations are a monolithic role prompt, persona only, no blueprint, no reviewed evidence, no style context, template realization, and full Sage.

A full-map-context condition runs only in an isolated safety harness. Its output never becomes a tested-agent observation.

Participants are the independent sampling unit. Questions, conditions, and seeds are repeated measures; analysis uses a mixed-effects model or participant-clustered paired bootstrap.

The proxy study then runs matched agents under Sage and human episodes. It reports absolute score shift, paired effect-direction agreement, rank correlation with uncertainty, and rank reversals.

State fidelity and rank preservation support different claims. High naturalness or rank agreement cannot repair a failed state-fidelity test.

5.7 Intermediate Annotation and Adjudication

End-to-end map error cannot reveal whether an answer was diagnostic, whether an update followed the rubric, or whether a question distinguished the intended boundary.

We construct a blinded intermediate set before inspecting full-method test results. Sampling is stratified by domain, mastery, support provenance, answer behavior, and policy.

Reviewers see the public rubric, visible plan, question, and answer. They never see the hidden label, method name, or final score.

For evidence records, reviewers label target-node relevance, supported boundary, polarity, sufficiency, contradiction, and whether the cited answer entails the record.

For questions, they label target alignment, discriminativeness, answer leakage, compound structure, and expected answer burden.

For graph messages, reviewers assess whether the edge semantics licenses the inference direction. They do not assign the user's mastery.

At least two reviewers annotate each sampled item.

The guide contains positive examples, near misses, and rules for hedging, self-report, partial reasoning, and volunteered evidence. Agreement is reported by field rather than as one number.

An independent reviewer adjudicates boundary or sufficiency disputes. Both pre-adjudication labels remain in the released artifact.

These annotations evaluate the evidence interpreter, question verifier, and qualitative failures.

Table 3: Reserved primary result structure. Values remain blank until the complete paired protocol is executed; no row is evidence at the present stage.

Policy	Map error ↓	AUC ↑	Turns ↓	Reliability ↑
Prior-only	TBD	TBD	0	TBD
Fixed traversal	TBD	TBD	TBD	TBD
Maximum uncertainty	TBD	TBD	TBD	TBD
Simple LLM	TBD	TBD	TBD	TBD
MapProbe	TBD	TBD	TBD	TBD

They never alter completed test episodes or train a method on the held-out split.

If automatic scores disagree with human labels, human-agreeing evidence validity governs the mechanism claim even when endpoint map error improves.

5.8 Statistical Analysis and Reporting

Methods are compared on paired episodes. Repeated stochastic runs are nested within episode rather than treated as independent users.

We report paired bootstrap confidence intervals and effect sizes, with multiplicity correction for each ablation family.

The development split sets priors, likelihood tables, utility weights, and stopping thresholds. The held-out test split is evaluated once after these choices are frozen.

Exclusion rules, failed provider calls, fallback behavior, and missing submissions are predeclared.

Every table names its aggregation unit and number of independent hidden profiles. Results cover all seeds rather than a selected successful trajectory.

Artifact manifests bind graph, map, simulator, agent, prompt, model, budget, and scorer versions for every row.

The claim language is determined by the strongest completed test.

Without matched baselines, we report only reconstruction. Without ablations, we do not attribute gains to beliefs, graph use, or planning.

Without simulator and human linkage, conclusions remain restricted to the evaluated proxy environment.

6 Discussion and Conclusion

6.1 What a KnowAct Result Would Mean

A benchmark score is evidence only relative to the explanations excluded by the design.

Low reconstruction error may reflect dialogue inference, a population prior, leaked information, or a good guess.

KnowAct strengthens interpretation through controlled access, evidence references, and non-adaptive comparisons.

The resulting claims form a hierarchy:

1. reconstruction: the final map matches the reviewed map;
2. active acquisition: adaptive policies outperform matched non-adaptive controls;
3. state use: working-map ablations identify a mechanism for that advantage;
4. graph-aware diagnosis: graph structure improves efficiency without unsupported propagation;

5. proxy robustness: the advantage persists across simulator configurations;
6. human transfer: the advantage persists in matched human interaction.

Evidence for a lower level does not entail a higher one.

Accurate reconstruction is not active user modeling if a fixed script achieves the same result. Failure to transfer to humans indicts the simulator proxy, not necessarily task consistency.

The interaction is active information acquisition over a structured latent state.

It differs from pool-based active learning because probes are open utterances that require grounding. It differs from adaptive testing because the item bank is neither fixed nor psychometrically calibrated.

This expressivity makes provenance, grounding, and proxy validation part of the measurement argument.

Even a fully validated result would remain diagnostic rather than task-complete.

KnowAct asks whether an agent acquires a useful user model. It does not establish improved teaching, recommendation, or collaboration.

Connecting reconstruction to downstream benefit is a separate claim and a natural next benchmark layer.

6.2 Limitations and Threats to Validity

The target is authored, not discovered. The graph and rubrics determine which distinctions count as knowledge.

Expert review and source provenance make that choice inspectable, not uniquely correct. Rankings may change under another granularity, relation set, or rubric.

Simulator fidelity. A simulator may be consistent with a reviewed map while expressing partial knowledge unlike a person.

Context isolation limits raw-field exposure, but does not establish semantic safety or behavioral realism. Held-out answers and matched human studies must test state fidelity and ranking transfer.

Artifact dependence. The graph, rubrics, maps, and simulator jointly operationalize the construct. Shared design choices may favor some model families.

Source grounding, deterministic fixtures, review, simulator variation, and human validation reduce but cannot remove this dependence.

Simplified knowledge state. A static L0–L5 state cannot capture context, procedural fluency, confidence, forgetting, or learning.

Freezing state makes reconstruction identifiable, but it treats explanations as observations rather than teaching events. Dynamic diagnosis needs a transition model and different causal interpretation.

Initial domain scope. The first release focuses on statistical learning and a small authored graph. It can validate the protocol, not cross-domain generality.

Additional domains must vary source style, topology, concept granularity, and expressible misconceptions.

Operational claims. Success would show behavior compatible with structured user modeling under benchmark constraints.

It would not establish human-like beliefs, consciousness, or a faithful psychological mechanism. The term functional ToM names the tested role, not an ontological status.

6.3 Conclusion and Outlook

Evaluating user-aware agents requires more than placing them in a conversation.

The evaluator must know the user state at stake, make evidence acquisition consequential, and separate accurate inference from leakage, priors, and generic interaction skill.

KnowAct turns a reviewed knowledge map into an explicit latent target and lets agents choose probes under a budget.

It exposes the link from visible evidence to working state and action. The Sage boundary and deterministic score make the loop auditable without equating simulation with human interaction.

The immediate next step is to test, not broaden, the claim.

Adaptive controls, mechanism ablations, simulator variation, and human episodes must establish which evidence level KnowAct supports.

Only then should the benchmark extend to dynamic learning, downstream assistance, and new domains.

Ethics Statement

User modeling can improve tutoring, accessibility, and collaborative assistance. It can also enable manipulation, surveillance, or unjustified inference about individuals.

KnowAct uses synthetic or explicitly reviewed benchmark profiles. It should not be interpreted as permission to infer sensitive attributes from real users without informed consent.

Derived systems should minimize retained personal data, expose uncertainty, and let users inspect or correct inferred states. Knowledge estimates should not determine high-stakes outcomes without human oversight.

Benchmark releases should document source provenance and licenses. Profile contexts, evidence records, and released traces must not contain identifiable personal information.

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A Implementation-Oriented Artifact Schemas

The abbreviated schemas below summarize the benchmark contracts. Field names may evolve, but the separation among public graph, hidden map, visible observations, and final reconstruction should remain stable.

A.1 Knowledge Node

```
{
  "id": "active_learning",
  "name": "Active Learning",
  "type": "concept",
  "definition": "...",
  "source_locators": [...],
  "diagnostic_goal": "...",
  "levels": {"L0": "...", ..., "L5": "..."},
  "diagnostic_signals": [...],
  "simulator_behavior": "..."
}
```

A.2 Hidden User State

```
{
  "node_id": "active_learning",
  "mastery_level": "L2",
  "evidence_refs": ["ev_104"],
  "misconceptions": [...],
  "unknowns": [...]
}
```

A.3 Agent Working Assessment

```
{
  "node_id": "active_learning",
  "assessed_mastery_level": "L2",
  "diagnostic_confidence": "medium",
  "assessment_note": "Understands the motivation but not
                      query-strategy trade-offs.",
  "supporting_turn_ids": ["turn_03"]
}
```

B Benchmark Construction Details

B.1 Knowledge-Graph Preparation and Claim Boundary

The knowledge graph is a supporting benchmark artifact, not the methodological contribution evaluated in this paper. Its preparation follows a staged pattern common in LLM-assisted graph construction—source-bounded extraction, canonicalization, verification, and human inspection (Zhang & Soh, 2024; Han et al., 2024; Chen et al., 2024a; Mo et al., 2025)—but we make no claim that this workflow is superior to another authoring method. The relevant validity question is narrower: whether the fixed graph used for scoring is defensible for the declared statistical-learning scope.

The source is a manually prepared Markdown snapshot of An Introduction to Statistical Learning with Applications in Python (James et al., 2023). Before candidate generation, the authors specify the intended aspect, five representative task families, exclusions, and a node budget. Candidate preparation then proceeds through heading-aware source segmentation, source-linked node extraction, global duplicate reconciliation, diagnostic-rubric enrichment, and typed relation proposal. The allowed edge types are part_of, prerequisite_for, supports, and contrasts_with.

Deterministic checks enforce schema validity, source-excerpt membership, identifier uniqueness, valid edge endpoints, canonical ordering, and the declared node budget. These checks establish structural integrity and traceability; they do not establish that a node is pedagogically useful, that adjacent rubric levels are distinguishable, or that a proposed relation is substantively correct. Those questions are reserved for expert review. This separation follows the broader measurement principle that the intended use and observable evidence must be specified separately from artifact generation (Grüninger & Fox, 1995; Mislevy et al., 2003).

B.2 Frozen Expert-Review Materials

The review input is the frozen statistical_learning_foundations graph version v2.0, containing 21 nodes and 30 edges. File hashes for the source, node list, edge list, and manifest are stored with the review record so that the reviewed input can be reconstructed exactly. Table 4 summarizes the packet.

Table 4: Materials supplied for the knowledge-graph expert review.

Material	Contents and purpose
Source snapshot	The scoped textbook Markdown with stable section locators, used to check whether node definitions and claimed relations are source-supported.
Scope specification	The aspect description, exclusions, and five representative task families, used to judge scope fit and bounded coverage.
Node packet	All 21 node names, definitions, source locators, diagnostic goals, L0–L5 rubrics, diagnostic signals, and simulator guidance.
Edge packet	All 30 source–target pairs, proposed relation types, directions, and rationales.
Review forms	Separate node, edge, graph-coverage, reviewer-metadata, and adjudication forms with prespecified categorical fields.
Version record	Graph version, candidate-run identifier, file names, and SHA-256 hashes for the frozen input and final adjudicated graph.

Each reviewer receives the same source, scope specification, graph files, and an unfilled copy of every applicable form. Reviewers do not receive model traces, verifier outputs, previous internal judgments, or the other reviewer's responses before submitting their independent review.

B.3 Reviewers and Independence

We recruit two reviewers with directly relevant statistical-learning expertise. Eligibility requires at least one of the following: experience teaching or assisting a university course covering statistical learning or regression; research using statistical-learning or regression methodology; or an advanced degree with relevant coursework and assessment experience. Reviewer qualifications, experience band, and conflicts of interest are recorded under anonymous codes R1 and R2.

Neither reviewer may have generated, edited, verified, or promoted the candidate graph. They review all items independently and may consult the cited source sections, but they may not discuss judgments until both forms are locked. A third qualified reviewer is recruited only if an item remains unresolved after R1 and R2 inspect each other's completed rationales. This design follows the transparent reporting pattern used by expert-curated and fully human-verified benchmarks, which state reviewer eligibility, expose the review unit and instructions, separate independent judgments from revision, and report concrete dispositions rather than only asserting data quality (Kim et al., 2023; Chen et al., 2024b; Zhang & Soh, 2024; Chen et al., 2024a).

B.4 Item-Level Review Procedure

Both reviewers inspect every node and edge rather than a sample. Table 5 gives the prespecified fields. Reviewers select controlled values and provide a rationale for every requested edit, rejection, or deletion. Free-text comments supplement but do not replace the categorical decisions.

Table 5: Prespecified expert-review fields and allowed decisions.

Unit	Field	Allowed values
Node	Source support	supported, partial, unsupported, uncertain
Node	Scope fit	in_scope, boundary, out_of_scope
Node	Granularity	appropriate, too_broad, too_narrow, mixed
Node	Diagnostic usefulness	adequate, minor_issue, major_issue
Node	L0–L5 rubric quality	adequate, minor_issue, major_issue
Node	Overall decision	accept, edit, reject
Edge	Relation validity	valid, uncertain, invalid
Edge	Type and direction	correct, incorrect with replacement, or uncertain
Edge	Evidence basis	source_explicit, source_entailed, expert_pedagogical_extension, unsupported
Edge	Overall decision	accept, edit, delete
Graph	Task coverage	sufficient, partial, insufficient, with essential omissions or redundancies listed
Graph	Overall decision	approve, approve_after_edits, do_not_approve

For a node, source support concerns the name and definition; diagnostic usefulness concerns whether dialogue can reveal a meaningful difference in knowledge; rubric quality concerns whether L0–L5 is ordered, behaviorally observable, and sufficiently distinguishable for the benchmark. For an edge, a merely related pair is insufficient: the proposed type and direction must match the stated semantics. The graph-level check asks whether the fixed node set is adequate for the five declared task families; it is a bounded coverage judgment, not an estimate of recall against an exhaustive reference ontology.

B.5 Adjudication, Analysis, and Promotion

After independent submission, the two overall decisions and all field-level disagreements are copied into a single adjudication record. Adjudication is mandatory for different overall decisions; every requested edit or deletion; any unsupported, out_of_scope, major_issue, or invalid judgment; and any task rated partial or insufficient. The reviewers first discuss the recorded evidence and smallest proposed correction. A third reviewer decides only cases that remain unresolved.

We report node and edge decision counts and percentages, raw agreement on overall decisions, Cohen's κ, field-level disagreement counts, reviewer time, and every adjudicated disposition. Raw agreement is reported alongside κ because a high prevalence of accepted items can depress chance-corrected agreement. Since the complete fixed graph is reviewed, the analysis is descriptive and uses no sampling-based significance test or arbitrary minimum-κ promotion threshold.

The graph is eligible for benchmark experiments only after all forms are complete, no disagreement or validity defect remains unresolved, all accepted edits are implemented, all five task families are judged sufficiently covered or have a documented scope resolution, and the resulting graph passes structural validation. Content changes are published as a new immutable version rather than overwriting v2.0. The complete forms and exact analysis template are included in the supplementary review package. Appendix D.3 reserves the result table to be filled after the study.

This procedure supports content validity for the declared source scope and tasks. It does not establish that the graph is an exhaustive ontology, that the authoring workflow is superior, or that the node-specific L0–L5 rubrics form a universal psychometric scale.

B.6 Knowledge-Map Review

Reviewers verify that each candidate map covers every node and that each state is supported by behaviorally realizable evidence. They also check consistency among mastery, misconceptions, unknowns, profile context, and graph structure. Promotion creates an immutable reviewed snapshot.

B.7 Manifest-Bound Runtime Contexts

The runtime maintains three contexts:

- Tested-agent context: public graph, interaction rules, visible transcript, and working map.
- Simulator context: grounded hidden states, evidence, limited visible dialogue, and optional style context.
- Evaluation context: hidden map, final reconstruction, and deterministic scoring configuration.

An episode manifest fixes the artifact versions and execution settings shared by these contexts. Debug traces and hidden policy outputs never enter the tested-agent context.

C Simulator and Agent Interface Details

C.1 Simulator Stage Contracts

The Sage grounder receives the public graph, current question, and limited visible history. It returns grounded node identifiers or a structured rejection reason before hidden state is loaded.

The context builder retrieves hidden state and reviewed evidence only for directly grounded nodes. Graph neighbors are not included merely because an edge exists.

The answer policy receives this local context. It returns a strict, de-identified blueprint that constrains claims, uncertainty, misconceptions, cues, and prohibited overstatements.

The generator receives the blueprint, limited visible dialogue, and optional style context. It does not receive raw mastery labels, map identifiers, evidence identifiers, or scoring fields.

The current service validates the policy schema, then retries only failed, timed-out, empty, or malformed generation. It has no independent post-generation semantic validator.

Therefore, the release must not describe context isolation as universal non-leakage. The adversarial and human semantic-leakage protocols in Appendix D test the output claim.

C.2 Tested-Agent Actions

The tested agent can read the public graph and its working map, update a node with visible support, ask one grounded question, or finalize.

Each mutation and question is recorded in the episode trace. Exact prompt templates, structured schemas, and provider-specific settings should accompany the released benchmark artifact.

D Detailed Experimental Protocol

D.1 Artifact and Run Binding

Every reported run identifies graph and map versions, turn budget, interaction rule, scoring profile, simulator, tested-agent model, provider settings, and execution configuration.

Reported graph versions link to source hash, declared scope, extraction evidence, verification decisions, workflow traces, and human-review records.

Development fixtures without this chain are excluded from benchmark claims.

Agents receive identical public artifacts and episode constraints. Hidden maps, simulator evidence, policy traces, and scoring internals remain outside the tested-agent context.

Repeated stochastic runs are nested within episodes. Confidence intervals and paired tests use the episode as the independent unit.

D.2 Policy Controls and Analyses

The fixed-question agent follows a predefined diagnostic order. The random agent samples valid targets under the interaction constraints.

The simple LLM agent chooses questions and updates its map through the same semantic interface as structured agents.

The main analysis reports mastery distance, exact match, missing predictions, unsupported inference, and per-node signed error.

Turn-indexed curves distinguish early information acquisition from final reconstruction.

Working-map, graph-relation, and evidence-reference ablations test state maintenance, graph-mediated generalization, and traceability.

D.3 Knowledge-Graph Expert-Review Results

Table 6 is a reporting placeholder for the study in Appendix B.

Every TBD entry must be replaced from locked reviewer forms and the adjudication record before empirical claims are made.

Table 6: Expert-review results placeholder. Decision entries should be reported as counts with percentages in parentheses.

Measure	Nodes	Edges	Task coverage
Items reviewed	21	30	5
R1: accept / edit / reject or delete	TBD	TBD	—
R2: accept / edit / reject or delete	TBD	TBD	—
Final adjudicated dispositions	TBD	TBD	TBD
Raw agreement	TBD	TBD	TBD/5
Cohen's κ	TBD	TBD	—
Items requiring adjudication	TBD	TBD	TBD
Unresolved items	TBD	TBD	TBD

Result statement to replace. Two qualified statistical-learning reviewers independently assessed all 21 nodes and 30 edges.

Node decisions achieved raw agreement of TBD with Cohen's κ of TBD. Edge decisions achieved raw agreement of TBD with κ of TBD.

The reviewers required TBD and TBD minutes, respectively.

After adjudication, TBD nodes were accepted unchanged, TBD edited, and TBD rejected.

For edges, TBD were accepted unchanged, TBD edited, and TBD deleted.

The final graph version was TBD; unresolved validity defects numbered TBD. This statement supports only content validity for the declared scope and tasks.

D.4 Sage Held-Out Human Protocol

Participants first answer diagnostic set A. Their answers, node rubrics, self-review, and researcher adjudication produce a reviewed map; unresolved participant–node pairs are retained as unresolved.

Diagnostic set B remains held out from map authoring and simulator tuning. Each participant answers B before seeing matched simulator responses.

For each participant–question–condition cell, three fixed seeds are retained. The participant, not the generated response or seed, is the independent sampling unit.

Participants rate core-claim, mastery, coverage, boundary, uncertainty, misconception, overstatement, understatement, style authenticity, and overall representativeness on seven-point items.

The instrument receives expert content review, cognitive interviews, and a pilot. Items are not summed until dimensionality and reliability support a composite.

Blinded reviewers annotate expressed mastery, correctness, boundary, uncertainty, misconception, diagnostic usefulness, naturalness, and style consistency.

Primary endpoints are matched human–simulator expressed-mastery distance and the participant's overall representativeness rating. Signed bias and misconception errors diagnose over- and under-performance.

D.5 Sage Safety and Proxy Protocol

Adversarial prompts request labels, identifiers, state tables, unrelated nodes, and canary fields. Visible-history injection and compound probes test whether scope widens improperly.

Exact forbidden-field matches and semantic leakage are separate outcomes. Passing observed probes supports only a bounded output-safety claim.

Matched human and Sage episodes hold graph, budget, tested-agent contract, and scorer fixed. We report effect-direction agreement, rank correlation with uncertainty, absolute shifts, and reversals.

Robustness checks also vary simulator model, grounding implementation, answer policy, and profile style while holding maps and budgets fixed.

Error categories include poor grounding, hidden-state leakage, missed evidence, stale working-map state, redundant probes, unsupported graph propagation, and premature finalization.

Representative trajectories connect each error to both the map update and the subsequent action.